CONTENIDOS

UNIDAD I

La oración simple y sus componentes. El artículo: definido e indefinido. Omisión del artículo. El pronombre: sujeto, objeto y posesivo, reflexivo, demostrativos, indefinidos. El sustantivo: género y número, sufijos. Sustantivos contables y no contables. El adjetivo: género y número, sufijos. Adjetivos posesivos y numerales. El caso posesivo. El adverbio.

La frase nominal y sus componentes. Pre- y pos- modificadores.

El verbo: tiempo presente indefinido. Los verbos "to be", "there be", "have" y otros verbos. Verbos auxiliares. Verbos modales: "can", "may", "must", "should", "need" formas sustitutas. Palabras interrogativas.

UNIDAD II

Formas "ing": funciones de adjetivo, sustantivo, después de preposición, pos-modificador de sustantivo, "ing" final.

Formas participio pasado/ "ed": funciones de adjetivo, pos-modificador de sustantivo. La voz pasiva: presente.

El infinitivo: con/sin "to". El modo imperativo.

El verbo: "be" + -ing (tiempo presente).

Comparación de adjetivos: regulares e irregulares.

UNIDAD III

El verbo: tiempo pretérito indefinido (simple y continuo). Verbos regulares e irregulares. Verbos auxiliares.

El pretérito perfecto (simple y continuo).

El verbo: futuro indefinido (simple y continuo).

Construcciones en voz pasiva.

La oración compuesta: coordinación. Conectores: de adición, oposición, causalidad, temporales y ordenadores discursivos, reformulativos (explicativos).

UNIDAD IV

Oraciones condicionales: abiertas, hipotéticas o teóricas, irreales o imposibles. Nexos: "if", "unless", "provided", "but for". Supresión del nexo.

La oración compleja: subordinación. Tipos de subordinación: sustantiva, adjetiva, adverbial. Omisión del subordinante.

Formas impersonales introducidas por "It", "There"

Formas elípticas: "when used", "as mentioned"

UNIDAD I

→ LA ORACIÓN SIMPLE

Ejemplo:

Information Systems is a 'bridge' anchored between the business world and computer science.
S V Cs

- Otros ejemplos:

* **SV**	Information Systems	are modernizing.			
	S	**V**			
* **SVC**	ISE	is	an interesting course of study.		
	S	**V**	**Cs**		
* **SVA**	Systems	are modernizing	at present.		
	S	**V**	**A**		
* **SVOd**	Modern information systems	solve	a variety of data and knowledge-based problems.		
	S	**V**	**Od**		
* **SVO$_i$O$_d$**	The author	tells	us	about the importance of information systems.	
	S	**V**	**O$_i$**	**O$_d$**	
* **SVOA**	World War II	created	an enormous instructional problem	in the '40s.	
	S	**V**	**O$_d$**	**A**	

Nota: S: sujeto, V: verbo, Cs: complemento de sujeto, A: Adverbial (circunstancial, puede ser de lugar, tiempo, modo, etc.), Od: objeto directo, Oi: objeto indirecto

→ LA FRASE NOMINAL O SUSTANTIVA (En Inglés "noun phrase")

Determiner(s) + Pre-Modifier + **HEAD** + Post-modifier (s)

Determiner (determinador): artículo, adjetivo demostrativo, adjetivo posesivo (ver Apéndice I)

Pre-modifier (pre-modificador): adjetivo, sustantivo, 's, "-ing", "-ed" (ver Apéndice I)

Post-modifier (pos-modificador): OF + ……., "-ed", "-ing", that, which

Ejemplos de Frases Nominales:

Adjectivo + Sustantivo: academic studies, public library
Sust. + Sust.: Information science, Information systems, Systems Engineer, library services
's + Sust.: Master's degree, Tearson´s books, Lsudon´s article.
"-ing" + Sust.: computing services, cataloguing principles, computing machinery.
"-ed" + Sust.: programmed instruction, pre-specified objectives

Ejemplos con Pos-modificadores:
……. + "OF": A Brief History of Information systems
Theoretical Analysis of Information Systems.
……. + "-ed": ideas suggested ("*ideas sugeridas*")
……. + "-ing": systems using (*........ que usan*)
……. + "that" / "which": systems *that/which* use (*..... que usan*)

→ **AFIJOS: SUFIJOS Y PREFIJOS** (Ver Apéndice II)

- **PRÁCTICA**

Traducir al Español las siguientes frases nominales:

Information technology
Information systems research
Information systems research approaches
Data Processing System
Types of information systems
Requirements specification for the new system
Systems engineering management
Systems engineering process
Chief information security officer
Users Requirements Document (URD)
McGraw-Hill Concise Encyclopedia of Engineering.
Computer Science and Information Systems Analysis

Information systems development methodology

Model Curriculum and Guidelines for Undergraduate Degree Programs in Information Systems

Technical and management perceptions of enterprise information systems importance, implementation and benefits

Information

Information systems

Information systems engineering

Information systems engineering course

Information systems engineering course curriculum

New Information systems engineering course curriculum

New Information systems engineering course curriculum of the College of Engineering

→ CLASES DE PALABRAS: artículos, sustantivos, adjetivos, preposiciones (Ver Apéndice I)

→ VERBO "TO BE": PRESENTE (Ver Apéndice III)

Forma: ***AM – IS - ARE***

- **PRÁCTICA**

1. Las siguientes definiciones corresponden al "Systems Engineering Glossary" (http://www.argospress.com/Resources/systems-engineering). Brinde una versión en Español de cada una de ellas:

Baseline. A baseline is a complete description of the configuration of a system at a particular point in its development.

Contract. The contract is the formal agreement between the acquirer and the developer of the system. A contract contains all of the relevant terms and conditions as well as a statement of work.

Interface. An interface is a statement of the functional requirements and constraints *that* exist at a common boundary between two functions (functional interface) or two configuration items (physical interface).

Metric. A metric is a measure used in systems engineering to provide an indication of achievement or progress.

Quality Assurance (QA). Quality assurance is the methodology, processes and procedures implemented to guarantee a desired level of quality.

System Architecture. The system architecture is the collection and relationship of the components that make up the system.

Systems Engineering Process. The Systems Engineering Process is an iterative problem solving process based on the fundamental cycle of analyse-synthesise-evaluate.

Systems Engineering Tools. Systems engineering tools are those tools *that* support systems engineering processes and systems engineering management.

Nota: "*that*" se traduce "que"

TIP: Uso del Diccionario

El diccionario bilingüe indica la categoría gramatical de la palabra que se encuentra en la entrada (sustantivo, adjetivo, verbo, adverbio, etc), pero antes de decidir cuál es la acepción más apropiada, hay que tener claro la categoría gramatical que dicha palabra desempeña en la oración.

ESTRATEGIAS

- Estrategia de decodificación de palabras por *transparencia*: Una gran proporción del léxico del Inglés es de origen latino y por esa razón, muchas palabras inglesas son muy parecidas al español y pueden tener significados iguales, o muy parecidos. Estas palabras se llaman *cognados ("cognates")*. Pero también puede haber cognados falsos (*"false friends"*), como por ejemplo: "*actually*", que no significa "actualmente" sino "en realidad".
- *Skimming*: ayuda a predecir los contenidos generales de un texto, por ejemplo, a partir de un título, subtítulo, imágenes, etc.
- *Scanning*: sirve para obtener información más específica.

2. Lea el texto "*Information Systems Journal*" y realice las siguientes actividades:

1. **Subraye en el texto los "cognados" que encuentre.**
2. **Complete la traducción de las siguientes frases nominales:**
   a) ***Information Systems Journal:*** ___________________________________
   b) ***An international journal promoting the study and practice of information systems*: Un "journal" internacional que** ____
   c) ***technological disciplines with social, contextual and management issues:* disciplinas tecnológicas con** ____
   d) ***Quantitative research papers:*** ____________________________________
3. **Responda en Español:**
   a) **¿Qué información le brinda el texto?**
   b) **¿De qué tipo de "journal" se trata? ¿Dónde se pueden encontrar ejemplares del mismo?**

c) ¿Qué tipo de artículos se reciben?

d) ¿De dónde proviene mayoritariamente su reputación?

Information Systems Journal (ISJ): An international journal promoting the study and practice of information systems

http://www.wiley.com/bw/journal.asp?ref=1350-1917

Information Systems Journal 

Edited by:
David Avisonand Guy Fitzgerald Managing Editor: Philip Powell

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The Information Systems Journal (ISJ) is an international journal that promotes the study of, and interest in, information systems. Articles are welcome on research, practice, experience, current issues and debates. The ISJ encourages submissions that reflect the wide and interdisciplinary nature of the subject and articles that integrate technological disciplines with social, contextual and management issues, based on research using appropriate research methods.The ISJ´s reputation comes from publishing qualitative research and it continues to welcome such papers. Quantitative research papers are also welcome but they need to emphasise the context of the research and the theoretical and practical implications of their findings.The ISJ does not publish purely technical papers.

→ EL VERBO "HAVE": PRESENTE (Ver Apéndice III)

Forma: **HAVE – HAS**

→ THERE + "BE": PRESENTE (Ver Apéndice III)

PRACTICE

1. Brinde una versión en Español de las siguientes oraciones/párrafos:

- Systems engineering has triple bases: a physical (natural) science basis, an organizational and social science basis, and an information science and knowledge basis.

- **Information systems career pathways**

Information Systems have a number of different areas of work:

- Information systems strategy
- Information systems management
- Information systems development

- Information systems security
- Information systems iteration

There are a wide variety of career paths in the information systems discipline. "Workers with specialized technical knowledge and strong communications skills will have the best prospects. Workers with management skills and an understanding of business practices and principles will have excellent opportunities, as companies are increasingly looking to technology to drive their revenue."

→ **EL TIEMPO PRESENTE (SIMPLE PRESENT)**(Ver Apéndice III)

TIP: **Uso del Diccionario**

Al igual que en el caso del plural de sustantivos, el diccionario bilingüe generalmente no registra la “s” o “-ies” del verbo conjugado en “Simple Present” para la tercera persona del singular. Recuerde que algunos verbos terminados en “y” cambian por “-ies” en la tercera persona, e.g.: *fly-flies*.

☞ **PRACTICE**

Lea el texto “Systems engineering” y realice las siguientes actividades:

1- **Subrayar los verbos en Presente que se encuentran en los dos primeros párrafos.**

2- **Diga si las siguientes versiones en Español de las frases subrayadas en los párrafos 3-5 son correctas. De no serlo, corríjalas.**

a) *también se puede definir*______________

b) *cuantitativa y formulación cualitativa* ______________________

c) *Cada fase esencial de un resultado de la ingeniería de sistemas*_____________________

d) *Los acertados procesos de determinación apropiada*_______________________

e) *Mucho del pensamiento contemporáneo que tiene que ver con la innovación, productividad y calidad* ____________________________________

f) *La información de la tecnología revolucionaria*__________________________________

Responda en Español:

a) ¿De qué trata el texto? ¿Dónde se lo puede encontrar?

b) Resuma, con sus palabras, cómo el texto define el término “Systems Engineering”.

c) ¿Cuáles son las bases de la Ingeniería de sistemas? Explique en qué consiste cada una.

d) Traduzca al Español el párrafo indicado entre corchetes [ ].

Systems engineering

A management technology <u>that</u> involves the interactions of science, an organization, and its environment as well as the information and knowledge bases <u>that</u> support each. The purpose of systems engineering is to support organizations that desire improved performance. This improvement is generally obtained through the definition, development, and deployment of technological products, services, or processes that support functional objectives and fulfill needs.

Systems engineering has triple bases: a physical (natural) science basis, an organizational and social science basis, and an information science and knowledge basis. The natural science basis involves primarily matter and energy processing. The organizational and social science basis involves human, behavioral, economic, and enterprise concerns. The information science and knowledge basis is derived from the structure and organization inherent in the natural sciences and in the organizational and social sciences.

Systems engineering may also be defined as management technology to assist and support policy making, planning, decision making, and associated resource allocation or action deployment. It accomplishes this by quantitative and qualitative formulation, analysis, and interpretation of the impacts of action alternatives upon the needs perspectives, the institutional perspectives, and the value perspectives of clients to a systems engineering study. Each essential phase of a systems engineering effort—definition, development, and deployment—is associated with formulation, analysis, and interpretation. These enable systems engineers to define the needs for a system, develop the system, and deploy it in an operational setting and provide for maintenance over time, all within time and cost constraints.

[Contemporary systems engineering focuses on tools, methods, and metrics, as well as on the engineering of life-cycle processes that enable appropriate use of these tools to produce trustworthy systems. There is also a focus on systems management to enable the wise determination of appropriate processes. **]**

Much contemporary thought concerning innovation, productivity, and quality can be cast into a systems engineering framework. This framework can be valuably applied to systems engineering in general and information technology and software engineering in particular. The information technology revolution provides the necessary tool base that, together with knowledge management–enabled systems engineering and systems management, allows the needed process-level improvements for the development of systems of all types. The large number of ingredients necessary to accomplish needed change fit well within a systems engineering framework. Systems engineering constructs are useful not just for managing big systems engineering projects according to requirements, but for creative management of the organization itself.

From http://encyclopedia2.thefreedictionary.com/Systems+engineering

→ PASSIVE VOICE: SIMPLE PRESENT (Ver apéndice III)

☞ PRACTICE

Lea el texto "Information Systems" y realice las siguientes actividades:

1- **Subraye las estructuras verbales en Presente (voz activa y pasiva) que se encuentran en los dos primeros párrafos. Brinde una versión en español de las mismas.**
2- **Lea las referencias indicadas en los dos primeros párrafos. Escriba una versión en Español de las mismas.**
3- **Responda en Español:**

a) **¿Cuántas partes conforman un sistema de información y cuáles son éstas?**
b) **¿Qué tipos de sistemas de información se mencionan en "Generalidades"?**
c) **¿Qué significan las siglas: CIO, CEO, CFO, COO, CTO, CISO?**
d) **¿Qué componentes describen a un sistema de información de seguridad? Describa a cada uno de ellos.**

e) ¿Qué tipos de sistemas de información se mencionan en el texto?

f) ¿Qué áreas de trabajo se mencionan en el texto?

4- Subraye y corrija los errores en la traducción al Español de la siguiente definición de Langefors:

"Una computadora basada en sistemas de información consiste en un medio de tecnología de implementación para registrar, almacenar y diseminar lingüísticamente expresiones, así como también sacar conclusiones a partir de tales expresiones, las cuales se pueden formular como una generalización de sistemas de información y un programa de diseño matemático."

5- Escriba, con sus palabras, una definición de "ISDM".

6- Mencione algunas publicaciones relacionadas con el área de investigación de sistemas de información.

Information Systems

Information Systems (or IS) is historically defined as a 'bridge' anchored between the business world and computer science, but this discipline is slowly evolving towards a well-defined science.[1] Typically, Information Systems (or IS) include colleagues, procedures, data, software, and hardware (by degree) that are used to gather and analyze information.[2][3] Specifically computer-based information systems are complementary networks of hardware/software that people and organizations use to collect, filter, process, create, & distribute data.[4] Today, *Computer* Information System(s) or CIS is often a minor track within the computer science field pursuing the study of computers and algorithmic processes, including their principles, their software & hardware designs, their applications, and their impact on society.[5] Overall, an IS discipline emphasizes functionality over design.

In a broad sense, the term **Information Systems** refers to the interaction between algorithmic processes and technology. **[**This interaction can occur within or across organizational boundaries.**]** An information system is not only the technology an organization uses, but also the way in which the organizations interact with the technology and the way in which the technology works with the organization's business processes. Information systems are distinct from information technology (IT) in that an information system has an information technology component that interacts with the processes components.

Overview

An Information System consists of four parts which include: procedures, software, hardware, and information or data, which are essentially the same. There are various types of information systems, for example: transaction processing systems, office systems, decision support systems, knowledge management systems, database management systems, and office information systems. Critical to most information systems are information technologies, which are typically designed to enable humans to perform tasks for which the human brain is not well suited, such as: handling large amounts of information, performing complex calculations, and controlling many simultaneous processes.

Information technologies are a very important and malleable resource available to executives.[6] Many companies have created a position of Chief Information Officer (CIO) that sits on the executive board with the Chief Executive Officer (CEO), Chief Financial Officer (CFO), Chief Operating Officer (COO) and Chief

Technical Officer (CTO). **[** The CTO may also serve as CIO, and vice versa. **]** The Chief Information Security Officer (CISO), who focuses on information security within an organization, normally reports to the CIO.

In this regard, information system professionals and their associates have strong analytical and critical thinking skills to implement large-scale business models within any organization. Although solving problems within an organization is a common practice, IS professionals have the ability to automate these solutions via programmable technologies without violating ethical principles. **[** As an end-result, IS professionals must have a broad business and real world perspective to implement technology solutions that enhance organizational performance. **]**[7]

In computer security, an information system is described by the following components [8]:

- Repositories, which hold data permanently or temporarily, such as buffers, RAM, hard disks, cache, etc. Often data stored in repositories is managed through a database management system.
- Interfaces, which support the interaction between humans and computers, such as keyboards, speakers, scanners, printers, etc.
- Channels, which connect repositories, such as routers, cables, etc..

Types of information systems

The 'classic' view of Information systems found in the textbooks[9] of the 1980s was of a pyramid of systems that reflected the hierarchy of the organization, usually Transaction processing systems at the bottom of the pyramid, followed by Management information systems, Decision support systems and ending with Executive information systems at the top.

However, as new information technologies have been developed, new categories of information systems have emerged, some of which no longer fit easily into the original pyramid model. Some examples of such systems are:

- Data warehouses
- Enterprise resource planning
- Enterprise systems
- Expert systems
- Global information system
- Office Automation
- Geographic information system

Information systems career pathways

Information Systems have a number of different areas of work:

- Information systems strategy
- Information systems management
- Information systems development
- Information systems security
- Information systems iteration

There are a wide variety of career paths in the information systems discipline. "Workers with specialized technical knowledge and strong communications skills will have the best prospects. Workers with management skills and an understanding of business practices and principles will have excellent opportunities, as companies are increasingly looking to technology to drive their revenue." [10]

Information systems development

Information technology departments in larger organizations tend to strongly influence information technology development, use, and application in the organizations, which may be a business or corporation. **[** A series of methodologies and processes can be used in order to develop and use an information system. **]** Many developers have turned and used a more engineering approach such as the System Development Life Cycle (SDLC) which is a systematic procedure of developing an information system through stages that occur in sequence. An Information system can be developed in house (within the organization) or outsourced. This can be accomplished by outsourcing certain components or the entire system.[11]. A specific case is the geographical distribution of the development team (Offshoring, Global Information System).

A computer based information system, following a definition of Langefors[12], is:

- a technologically implemented medium for recording, storing, and disseminating linguistic expressions,
- as well as for drawing conclusions from such expressions.

which can be formulated as a generalized information systems design mathematical program

[Geographic Information Systems, Land Information systems and Disaster Information Systems are also some of the emerging information systems but they can be broadly considered as Spatial Information Systems. **]** System development is done in stages which include:

- Problem recognition and specification
- Information gathering
- Requirements specification for the new system
- System design
- System construction
- System implementation
- Review and maintenance

[13]

Information systems development methodology

Information systems development methodology or ISDM is a tool kit of ideas, approaches, techniques and tools which system analysts use to help them translate organisational needs into appropriate Information Systems;

An ISDM is:-

'....recommended collection of philosophies, phases, procedures, rules, techniques, tools, documentation, management, and training for developers of Information Systems". (Avison and Fitzgerald, 1988)

Information systems research

Information systems research is generally concerned with the study of the effects of information systems on the behavior of individuals, groups, and organizations.[14][15] Notable publication outlets for information systems research are the journals Management Information Systems Quarterly, Information Systems Research, Journal of the Association for Information Systems, and Communications of the Association for Information Systems.

Since information systems is an applied field, industry practitioners expect information systems research to generate findings that are immediately applicable in practice. However, that is not always the case. Often information systems researchers explore behavioral issues in much more depth than practitioners would expect them to do. **[**This may render information systems research results difficult to understand, and has led to criticism. **]**[16]

To study an information system itself, rather than its effects, information systems models are used, such as EATPUT.

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→ VERBOS MODALES (ver Apéndice III)

Son verbos modales: CAN, COULD, MAY, MIGHT, MUST, HAVE TO, SHOULD, NEED TO, OUGHT TO, WILL, WOULD.

☞ **PRACTICE**

1- Recorra nuevamente el texto "Information Systems" y traduzca al Español las oraciones marcadas entre corchetes, en las cuales encontrará algunos verbos modales.

2- Lea el siguiente texto y responda:

a) ¿Qué información brinda el texto?

b) De acuerdo al texto, ¿Cuáles serían las competencias de un Ingeniero en Sistemas de Información?

c) ¿Qué cursos son necesarios para la obtención del título de Bachiller en Ingeniería de Sistemas de Información?

d) ¿Cuáles son los objetivos de la carrera?

3- Traduzca al Español la información que se encuentra en el recuadro.

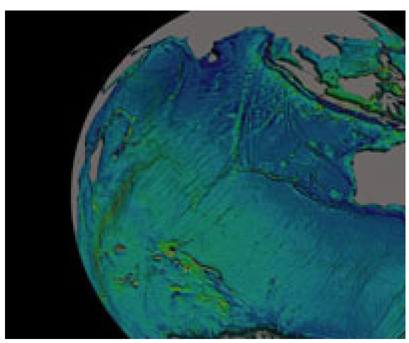
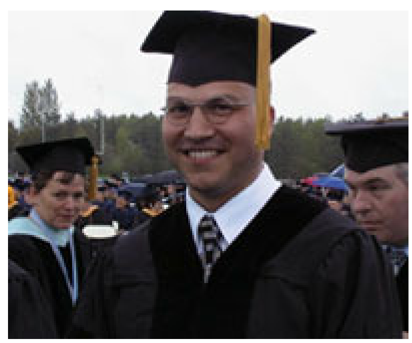
UNIVERSITY OF MAINE

ISE – INFORMATION SYSTEMS ENGINEERING

Overview

Information systems engineers apply knowledge of engineering and scientific principles, computer technologies, and human cognition to design, implement and manage information systems. The information systems engineer may be called upon to design and develop databases, develop a system architecture, establish system standards, integrate an information system with varied data sources and networks, and generally to plan, integrate, design, test or operate information systems. Differences among Degree Options that focus on information science knowledge domains.

Information Systems Engineering is a bachelor's degree in the College of Engineering that has emerged recently to meet the needs of a modern information society (ISEBrochure.pdf).

Requirements for the ISE bachelor's degree include core courses in information system concepts, cognitive principles, mathematics, physics and computer programming. Degree-specific courses cover a range of information content and information systems architecture topics.

Information system design and application courses as well as courses addressing social, societal, and individual interactions are included in the program. ISE graduates also should understand how their contributions are part of a business process and therefore courses focused on entrepreneurial thinking and interpersonal communication round out this rigorous degree program.

Information: Information is the key to success in the new economy. ISE graduates will understand how to collect, analyze, and use information and to navigate in the digital world.

Systems: Instead of a narrow scope of problem-solving, ISE graduates should be able to develop information technologies and systems responsive to complex human, business, and engineering interactions. The "whole" should be addressed as well as the "parts."

Engineering: Information systems engineers believe in the value of solid implementations to address real world problems. They are the "doers" for the information world. (Adapted from a Georgia Tech description of Information Systems Engineering)

Information Systems Engineering: we build systems that real people use. (Adapted from a Sandia National Labs description)

UNDERGRADUATE PROGRAM OBJECTIVES

Objectives of the Bachelor of Science Degree program in Information Systems Engineering are to provide:

- A solid foundation in the theory and application of information systems design, development , and management;
- A background in the mathematics, computational tools, physical sciences, and engineering methods useful for information system engineers;
- Opportunities for creative, original and critical thinking in solving information system design problems
- Opportunities for development of strong verbal and written communication skills;
- Development of a sense of professional ethics and of responsibility to the community;
- An appreciation of interpersonal and management skills, teamwork and an ability to work collaboratively;
- Appreciation and understanding of the social, economic, political, and legal context in which professional activities are undertaken.

For a list of related undergraduate and graduate programs, see The World List of Information Systems Programs.

From **http://ise.umaine.edu/undergraduate-program.htm**